

Demystifying Brazilian Ecommerce

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Analysis Objective

1

Understand what drives sales

Geography, payment method, and behaviour

2

Identify most profitable products

Revenue vs units and category outliers

3

Characterise top spenders

Who they are and what sets them apart

4

Provide solutions from insights

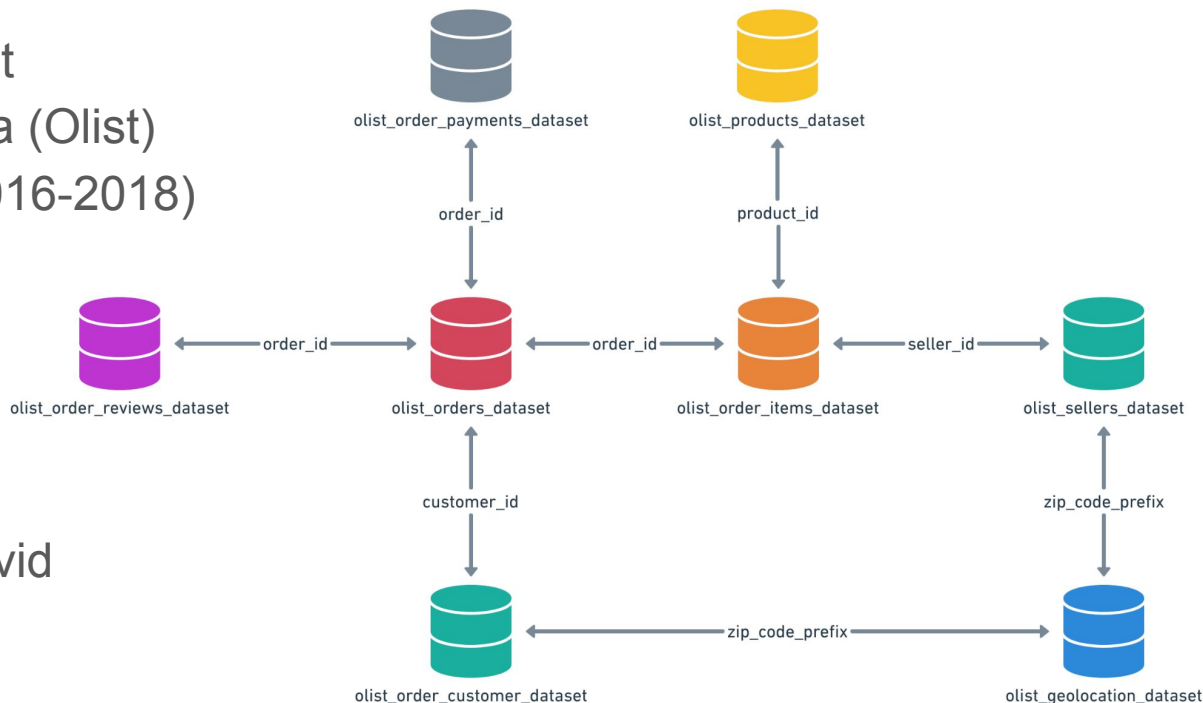
Three actionable recommendations

About the data



About the data

- Public Kaggle dataset
- Real commercial data (Olist)
- Over 100k orders (2016-2018)



Data limitations:

- limited timespan
- doesn't consider Covid

Current sales drivers



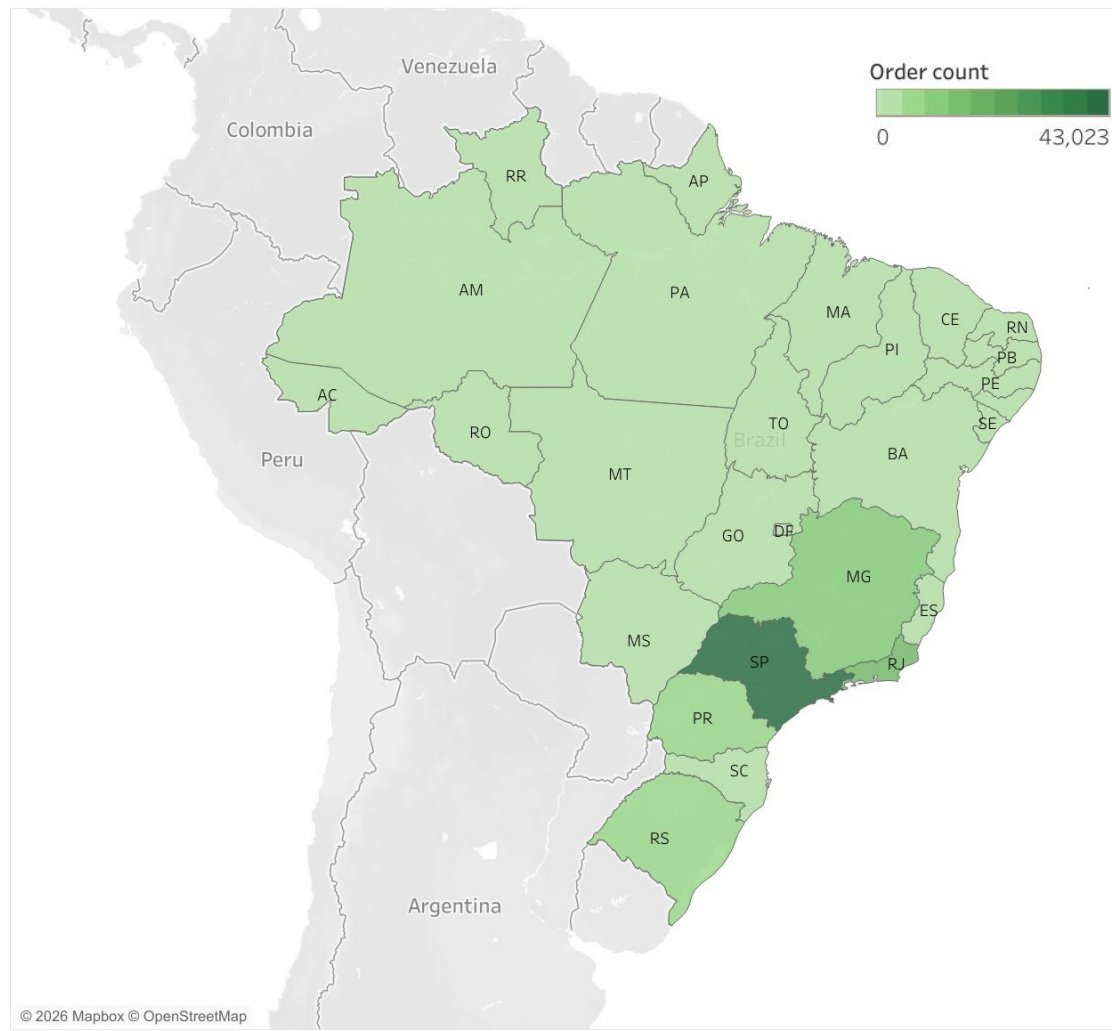
How to define sales?

- Sales defined as **total revenue** earned (product/category)
- Important customer shopping **context**
- Analyse surrounding factors (payment method, no. of purchases)
- All financial data in **R\$**

Current sales drivers

Shopping context

- Map shows order data by state
- **São Paulo clear outlier**
- Order count affected by state GDP



Current sales drivers

Shopping context

- Map shows Nominal GDP based on state
- Confirms conclusion from previous slide

Map of the Brazilian states by nominal GDP



What drives sales?

Shipping statistics

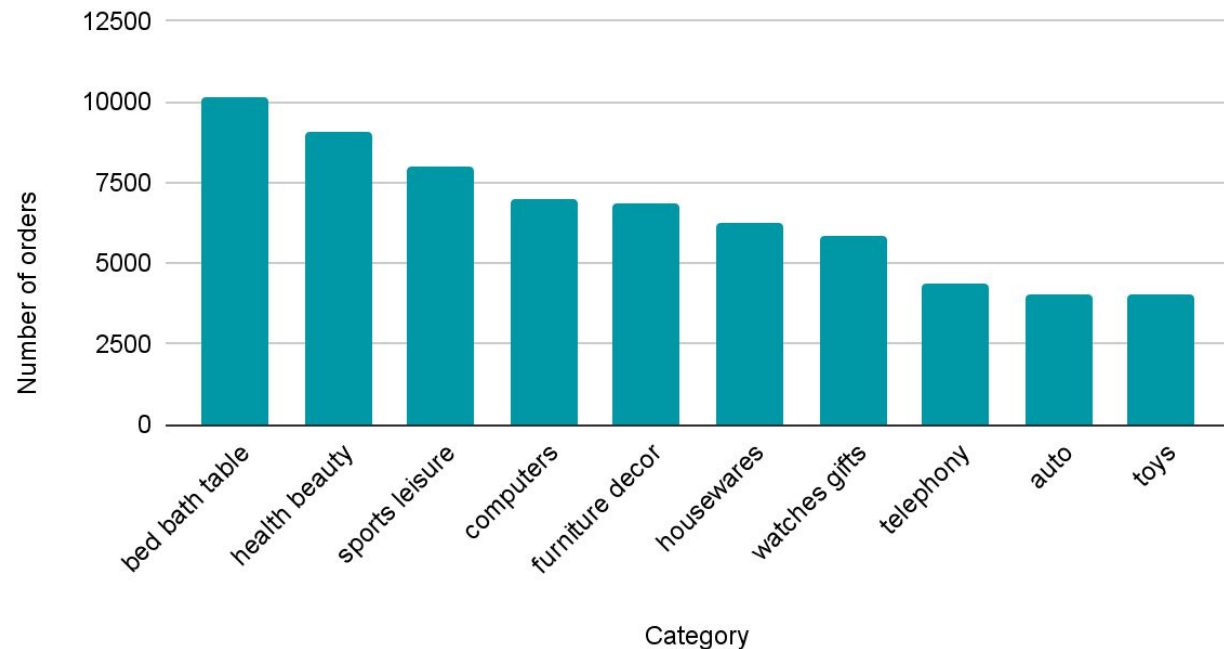
Delivery rate at 97.3% vs 95% Industry average
Suggests high consumer satisfaction

What drives sales?

Orders by category

- 72 unique categories
- Top 10 selected

Number of orders in each category (top 10)



Identifying profitable products

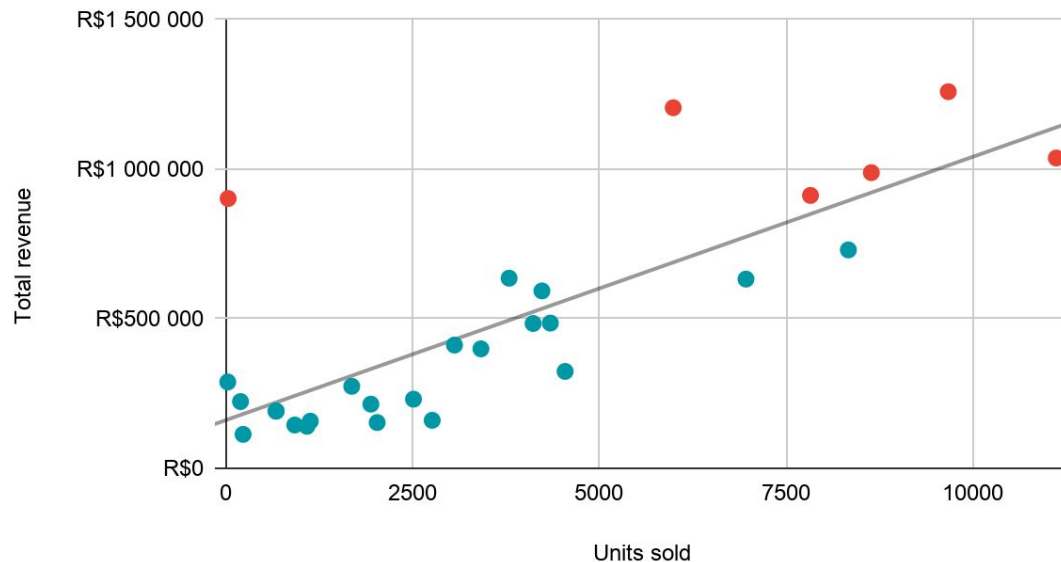


Identifying profitable products

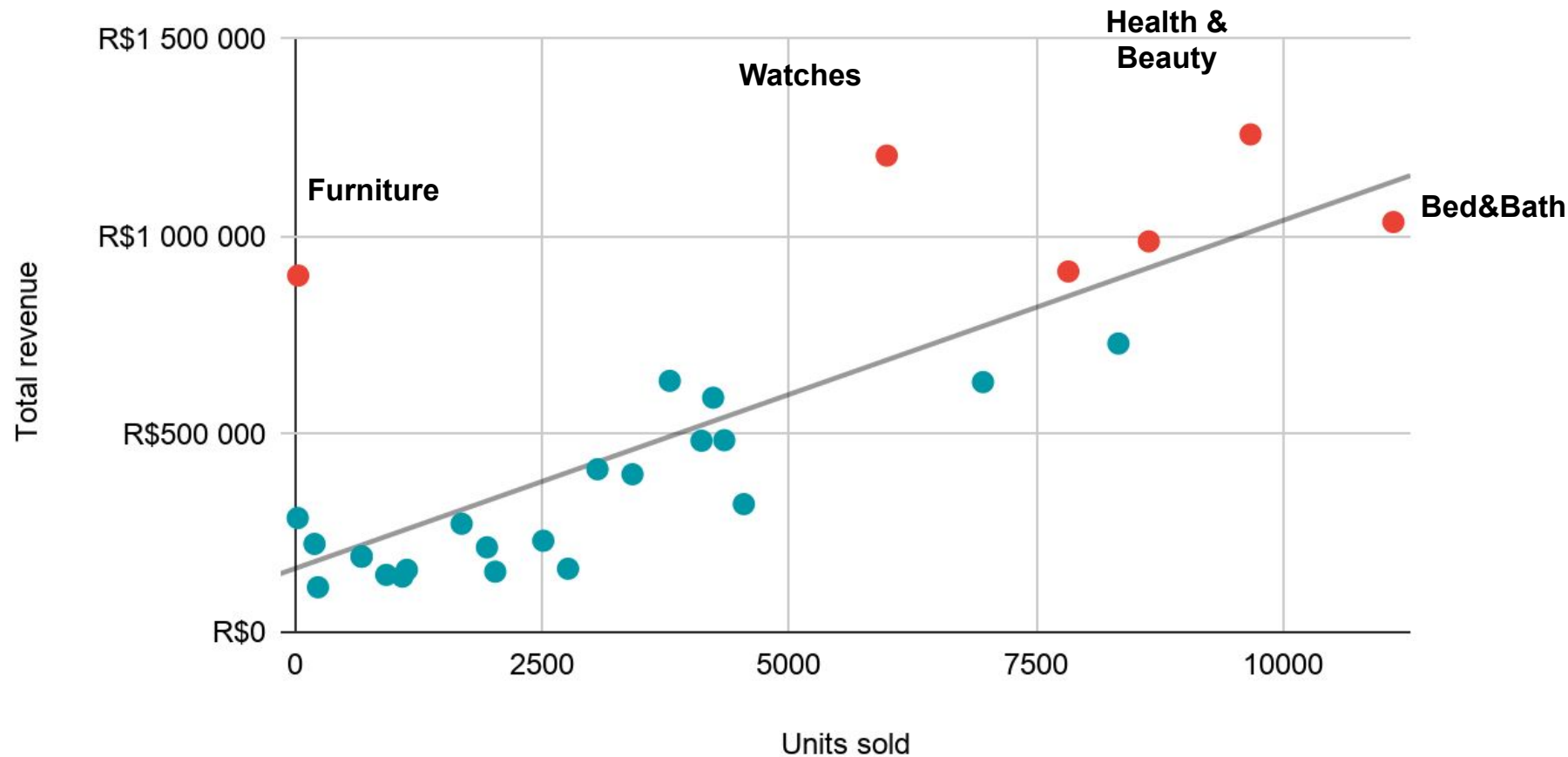
Category vs Units sold

- Data filtered for readability
- category predicts ~64% of the variation in sales volume ($R^2 = 0,64$)

Total revenue to Units sold by Category



Total revenue to Units sold by Category: Key Categories

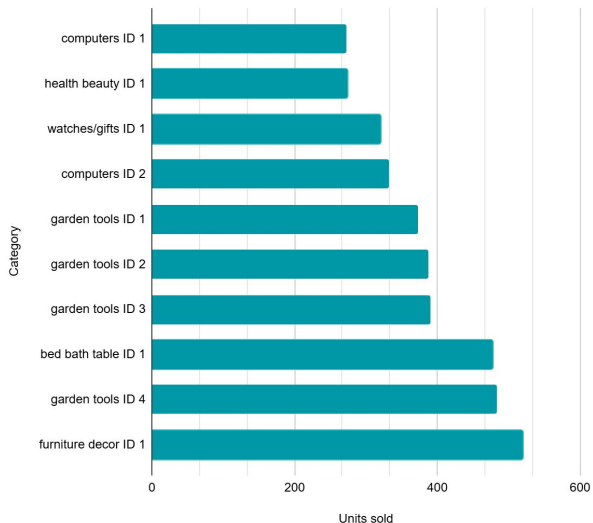


Identifying profitable products

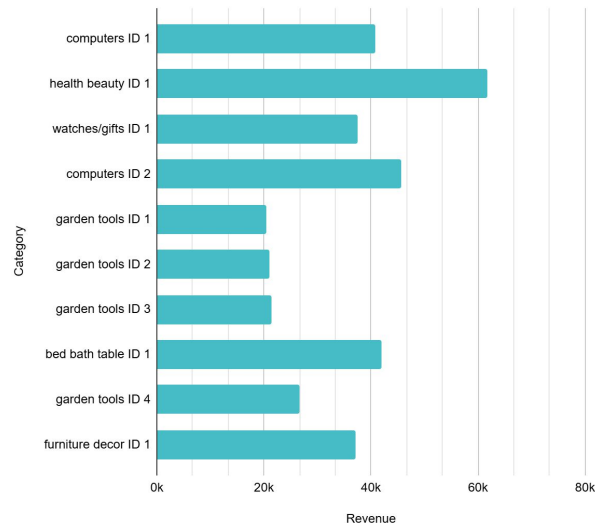
Top selling products by category

- **Garden tools:** majority driver of units sold
- **computers & watches** provide the biggest **profit margins**
- real product ID unavailable

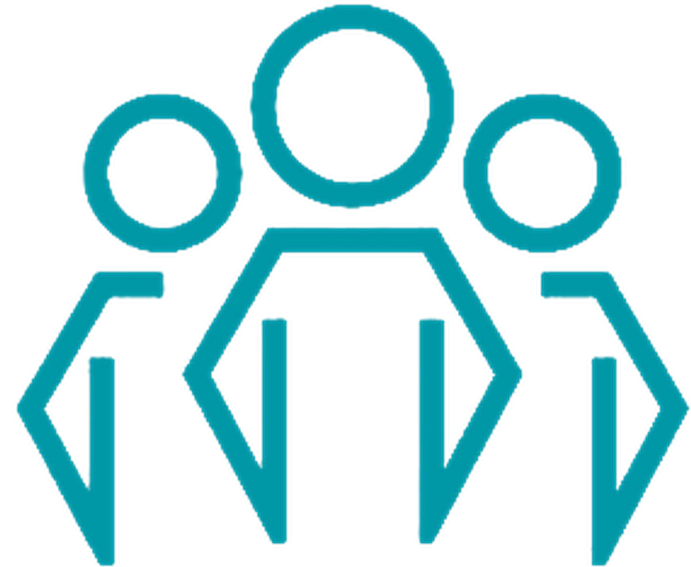
Units sold by Product category (Top 10)



Revenue by Product category (Top 10)



What sets top spenders apart?

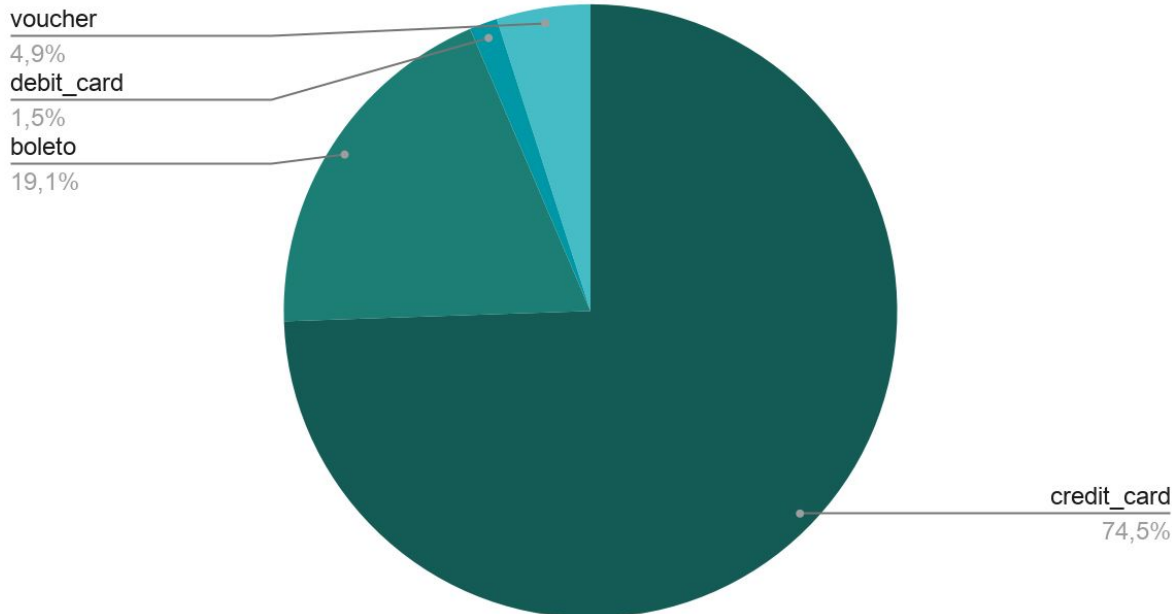


What sets top spenders apart?

The average customer

- AVG order value:
R\$156,18
- AVG no. of orders:
1,03

Payment Method



What sets top spenders apart?

Top 5 customers by spending

- majority purchases via credit cards
- **no debit card use**

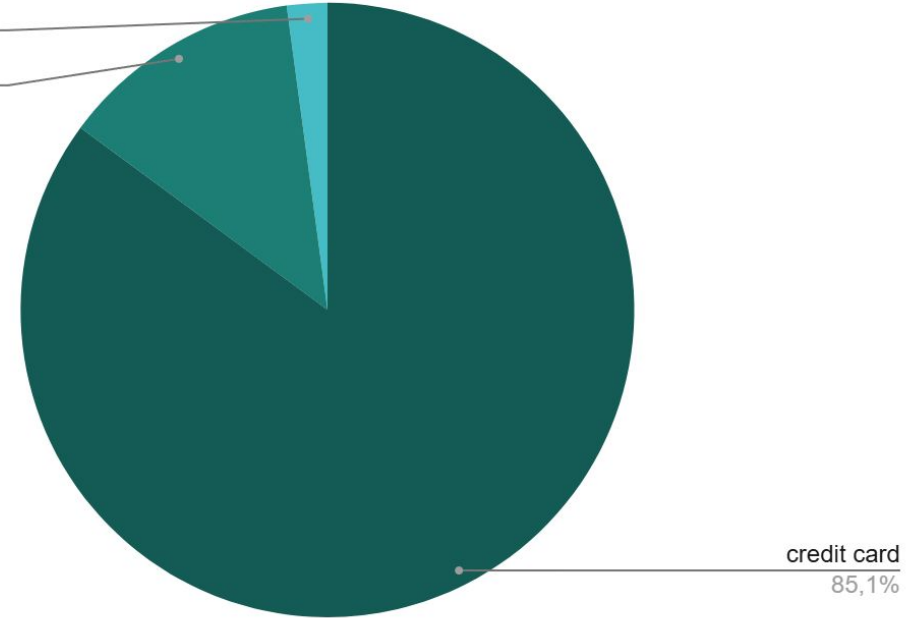
Top 5 customers by payment type

voucher

2,1%

boleto

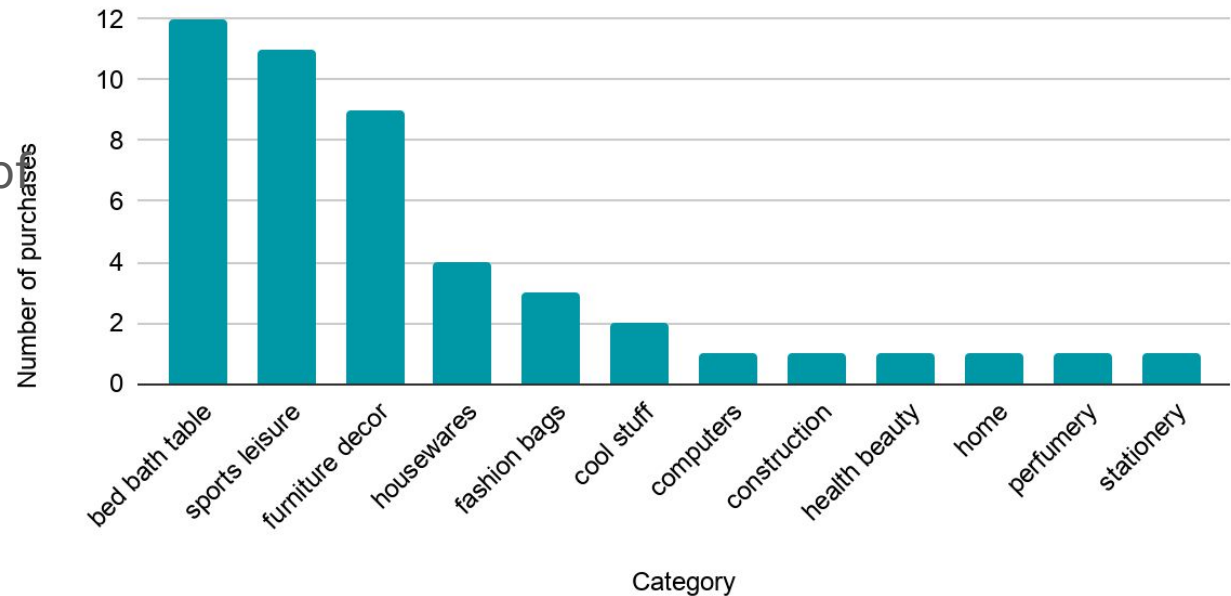
12,8%



What sets top spenders apart?

- AVG order value: **R\$110,59**
- AVG no. of orders: **9,4**
- Top spenders valuable because of **high order frequency**

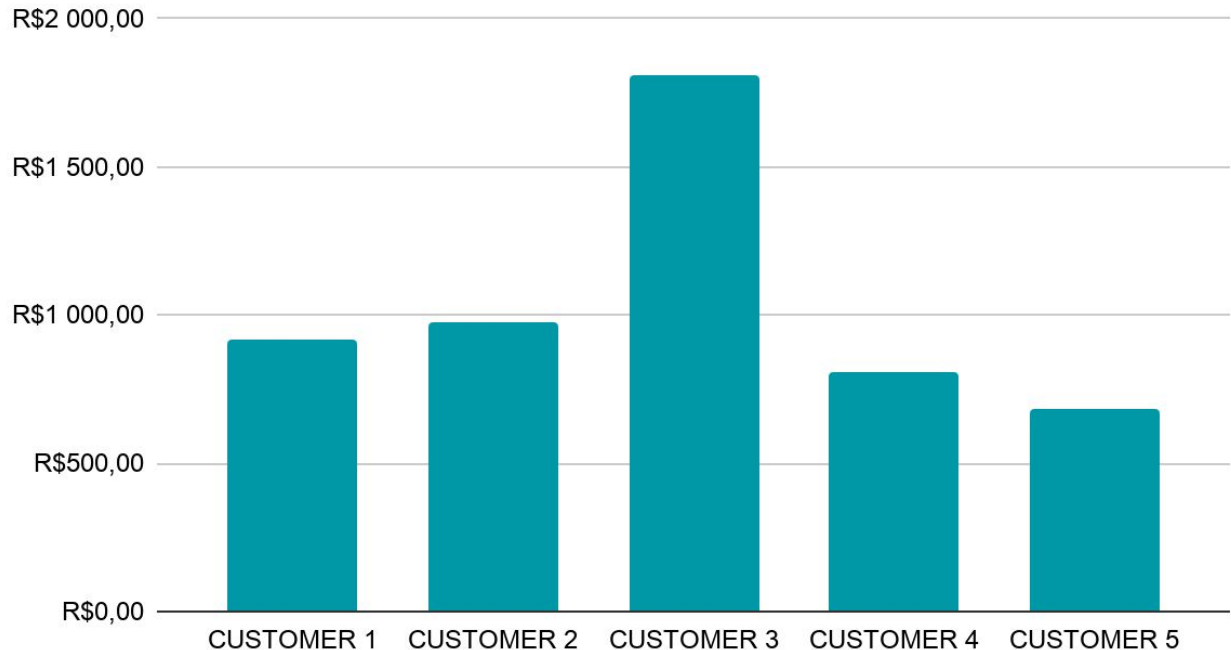
Number of purchases in each category made by Top 5 customers



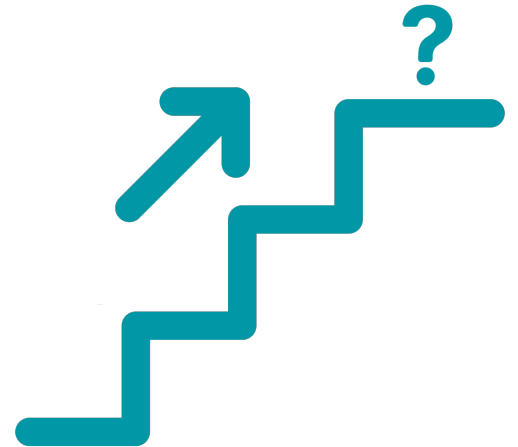
What sets top spenders apart?

- Top spenders generate close to **R\$1000** in profit (equivalent to **9 single time shoppers**)

Total revenue generated by the Top 5 customers in Brazilian Reals



What can be done?



Sales improvement suggestions

Option 1

Improving High-end sales

Improve functionality and platform support towards **high ticket orders**

Option 2

Adjust payment methods

Provide enticement for users to use **boleto**. Improves customer lifetime value and encourages more single purchase customers.

Option 3

Customer reward system

Honor returning customers by providing a **small benefit**.
Award new customers by giving a **first-time bonus**

Conclusions

01

Ecommerce
location closely
tied to state GDP

02

Stellar delivery
rate signals
high-quality
service and
products

03

Acquiring new
customer bases is
crucial to further
scaling

04

Introducing
multiple
incentives should
prove effective

Thank You

Szymon Rosiński

Data Analysis | Brazilian Ecommerce (Olist, 2016–2018)

Questions? See [Appendix slide](#) for anticipated Q&A

Appendix and Q&A

